9.5 Sink or Float? Relative Density

4th Grade Science · Video: Sink or Float? (SciShow Kids)

Name: _____ Date: _____

Watch the video and fill in the blanks as you go.

Making Predictions

Some things sink right to the bottom, while other things _____ on the top.

You can learn a lot about something by seeing what it does in the _____.

Before testing, the team guessed whether each object would sink or float and kept track on a _____.

The Test

The objects were made of different materials, like metal, _____, and wood.

The stick _____.

The metal spoon _____.

The plastic spoon is the same shape as the metal spoon, but it _____.

The penny is small and light, but it _____.

The metal spoon and the stick both felt heavy, but only the stick floated. So sinking or floating is not just about how _____ something feels.

What Is Density?

Everything is made of tiny _____ that we can't see.

How _____ an object is depends on how much space is between its particles.

Particles packed close together, like marbles right next to each other, make an object _____ dense.

Particles spaced far apart make an object _____ dense.

The metal spoon is _____ dense than the plastic spoon.

Sink or Float?

Everything has density, even _____!

If something is more dense than water, it _____.

If something is less dense than water, it _____.

The metal spoon and the penny were more dense than water. The stick and the plastic spoon were _____ dense than water.

Will it sink or float? It depends, in part, on _____.

Vocabulary

Word	Meaning
particle	a tiny piece of matter that is too small to see
density	how much matter is packed into a space; how close together the particles are
sink	to drop to the bottom of the water
float	to stay on top of the water
relative density	whether something is more dense or less dense than water

Key Takeaways

• Objects that are more dense than water _____. Objects that are less dense than water float.

• Sinking or floating in water is a property we can use to sort and classify matter.